

## Linear Algebra – Problem Set 2

*Row Echelon Form & the Determinant · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)*

### Problem 1

Compute  $\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$  by reducing to row echelon form. Use row replacements only — no swaps, no scalings — and state why your answer is legitimate.

### Problem 2

Compute  $\det \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$  by reducing to row echelon form. A leading entry of zero forces a swap. Perform the swap, finish the reduction, and explain exactly where the swap's effect on the determinant enters your answer.

### Problem 3

Let  $A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ . Without computing the determinant, use row echelon form to decide whether  $\det A$  is zero, and say which single observation settles it.

### Problem 4

A common habit is to normalize each row while reducing (dividing the pivot row by its leading entry). Explain why this is a bad idea when the goal is the determinant. Give a  $2 \times 2$  example making the cost concrete, and say which single step type you should avoid.

### Problem 5

Starting from  $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 6 & 5 & 1 \end{bmatrix}$ , the following operations are performed in order:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1, \quad R_3 \leftarrow R_3 - 2R_2,$$

producing an upper triangular matrix  $U$ . Find  $\det A$  *without* redoing the elimination.

### Problem 6

Let  $A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 1 & 3 \end{bmatrix}$ . Reduce to row echelon form. Then (a) report the rank, and (b) exhibit the exact relation among the rows that forces  $\det A = 0$ .

**Problem 7**

Let  $A_t = \begin{bmatrix} t & 1 & 0 & 0 \\ 0 & t & 1 & 0 \\ 0 & 0 & t & 1 \\ 1 & 0 & 0 & t \end{bmatrix}$ . Use row echelon form to find all *real* values of  $t$  for which  $\det A_t = 0$ . Justify that your list is complete.

**Problem 8**

Compute  $\det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{bmatrix}$  by row echelon form. Fractions are unavoidable here; carry them exactly rather than rounding.

**Problem 9**

Compute the determinant of  $A = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 2 & 4 & 1 \\ 1 & 4 & 4 & 5 \end{bmatrix}$  by reducing to row echelon form. Find a reduction that needs neither swaps nor scalings, and say why such a reduction exists here at all.

**Problem 10**

Let  $B_t = \begin{bmatrix} 1 & 1 & t \\ 0 & 1 & 1 \\ 1 & 0 & t \end{bmatrix}$ . Show by row reduction that  $\det B_t = 1$  for *every*  $t$ . Then explain why expanding along the first row would have made this much harder to see.

**Problem 11**

Let  $A$  be  $n \times n$ . Prove that  $\det A \neq 0$  if and only if the row echelon form of  $A$  has a pivot in every row. (Your proof may use the three determinant effects of elementary row operations, but it must not use the general theory of rank.)

**Problem 12**

True or false? Justify each answer (if false, give a counterexample or correct the statement).